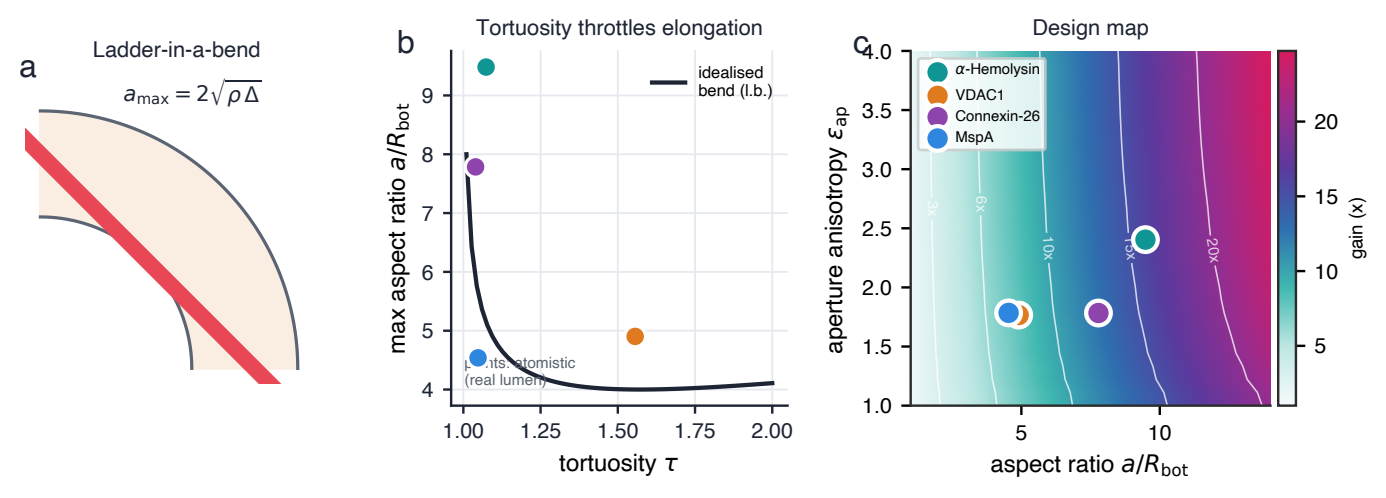

a
Ladder-in-a-bend
$a_{max} = 2\sqrt{\rho\,\Delta}$
b
Tortuosity throttles elongation
idealised bend (l.b.)
max aspect ratio a/R_{bot}
tortuosity τ
1.00
1.25
1.50
1.75
2.00
4
5
6
7
8
9
(real lumen)
c
Design map
α-Hemolysin
VDAC1
Connexin-26
MspA
aperture anisotropy ε_{ap}
aspect ratio a/R_{bot}
gain (x)
3x
6x
10x
15x
20x
1.0
1.5
2.0
2.5
3.0
3.5
4.0
5
10
15
20